

ACTIVE INGREDIENTS

MICROBES IN FOOD

Microorganisms are tiny life forms not visible to the naked eye, and sometimes called microbes. While some have a bad reputation for causing disease, many of our favorite foods owe their flavor and texture to microorganisms. Microbes such as bacteria, yeast, or mold can change the chemistry of food in a natural process called fermentation. People have fermented food for thousands of years to make it last longer, taste better, or provide more nutrition.

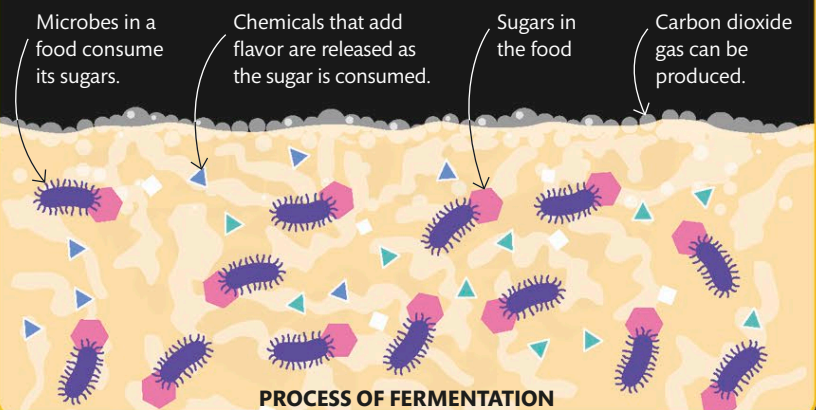


ADDING BACTERIA

In an early stage of the cheese-making process, cheese makers mix a yellow powder containing bacteria into vats of milk. Bacteria eat the energy-rich sugar inside milk and convert it into substances that allow the milk to harden.

FERMENTATION

Fermentation is a natural chemical reaction that happens when microbes consume food. Different kinds of microbes cause different types of fermentation. The creatures consume sugars inside the food—milk, in the case of cheese—and then convert them into substances that add flavor. In cheese, microbes eat the sugars and produce lactic acid, which not only flavors the cheese but also helps it last longer.



Aging cheese

At a cheese aging cellar in France, a cheese maker taps the wheels of cheese to check how ripe they are. Some types of strong-tasting cheese are left to ripen for several years. This allows bacteria and mold to continue fermenting the cheese so that its rich flavors and textures continue to develop.